

Guess The Number Game

Introduction

The **Guess the Number Game** is a simple Python-based number guessing game. The program randomly selects a number between any random range and the player has a limited number of attempts to guess it correctly. The game provides feedback on whether the guess is too high or too low and prevents duplicate guesses.

In this game, python fundamentals like if, elif, while loop, user input are used.

Features

1. Randomly selects a secret number range.
2. Provides feedback on whether the guess is too high or too low.
3. Prevents duplicate guesses.
4. Input validation to ensure only valid numbers are entered.

Code Explanation

Dependencies

This script uses Python's built-in **random** modules, so no additional dependencies are required.

Importing Required Libraries

```
import random
```

Initialization of Game

```
print("Welcome to the Number Guessing Game!")
lower = int(input("Enter the lower bound: "))
upper = int(input("Enter the upper bound: "))
max_attempts = int(input("Enter the maximum number of attempts allowed: "))
```

Generating the Random Number

```
number_to_guess = random.randint(lower, upper)
```

Providing Feedback

```
if guess < number_to_guess:
    print("Too low! Try again.")
elif guess > number_to_guess:
    print("Too high! Try again.")
else:
    print(f"Congratulations! You guessed the number in {attempts}
attempts.")
    return
```

Game Over Condition

```
print(f"Sorry, you've used all {max_attempts} attempts. The number was
{number_to_guess}.")
```

Save the code

Save the code with name followed by **.py** extension

guess_num.py

Run the code

guess_num.py

Example Output

```
PS E:\python> & C:/Users/KHATRI/AppData/Local/Programs/Python/Python314/python.exe
e:/python/guess_num.py
Welcome to the Number Guessing Game!
Enter the lower bound: 1
Enter the upper bound: 10
Enter the maximum number of attempts allowed: 3
Enter your guess (Attempts remaining: 3): 5
Too low! Try again.
Enter your guess (Attempts remaining: 2): 8
Too low! Try again.
Enter your guess (Attempts remaining: 1): 9
Too low! Try again.
Sorry, you've used all 3 attempts. The number was 10.
```

```
PS E:\python> & C:/Users/KHATRI/AppData/Local/Programs/Python/Python314/python.exe
e:/python/guess_num.py
```

Welcome to the Number Guessing Game!
Enter the lower bound: 1
Enter the upper bound: 20
Enter the maximum number of attempts allowed: 6
Enter your guess (Attempts remaining: 6): 15
Too high! Try again.
Enter your guess (Attempts remaining: 5): 10
Too high! Try again.
Enter your guess (Attempts remaining: 4): 9
Congratulations! You guessed the number in 3 attempts.
PS E:\python>

Possible Enhancements

GUI with Tkinter for better user experience.